Classical Field Theory — Exercise Sheet 15

Beyond the one-dimensional cubic proof

Solutions

Use the mass-one Klein–Gordon group $U_d(t)$ on $\mathcal{E}_d = H^1(\mathbb{R}^d) \times L^2(\mathbb{R}^d)$, with Hilbert norm $\|(f,g)\|_{\mathrm{lin}}^2 = \|f\|_{H^1}^2 + \|g\|_2^2$. Lecture 15 proves the forced estimate in one dimension; Problem 2 checks its dimension-independent Fourier proof. All fields below are real.

Exercise 15.1 (A general potential and an affine background). Let $V \in C^3(\mathbb{R})$ and let b be a smooth bounded solution of $-b'' + V'(b) = 0$ on $\mathbb{R}$. Do not assume $b \in L^2$. For $\phi = b + v$ consider $\phi_{tt} - \phi_{xx} + V'(\phi) = 0$.

(a) Derive the mass-one perturbation force $N_b(v)$, and show that it maps $H^1(\mathbb{R})$ locally Lipschitzly to $L^2(\mathbb{R})$, vanishing at zero. Give a constant on a ball of radius R using the maximum of $|V''|$ on a specified compact interval.

(b) For every initial ball $\|(v_0, g_0)\|_{\mathrm{lin}} \le B$, $B \ge 1$, give a common contraction radius and positive time. Prove existence, uniqueness among weak energy solutions, continuous dependence, and the finite-lifespan alternative. Which derivatives of V did the estimates use?

Solution. For the first part, cancellation of $-b'' + V'(b)$ gives

$$v_{tt} - v_{xx} + v = N_b(v), \qquad N_b(v) = v - [V'(b+v) - V'(b)], \qquad N_b(0) = 0.$$

For $\|v\|_{H^1}, \|w\|_{H^1} \le R$, the one-dimensional embedding bounds both perturbations by $R/\sqrt{2}$. Their background-shifted values lie in $[-M, M]$, where $M = \|b\|_\infty + R/\sqrt{2}$. The scalar identity

$$V'(b+v) - V'(b+w) = (v-w) \int_0^1 V''(b + w + \theta(v-w))\mathrm{d}\theta$$

gives $\|N_b(v) - N_b(w)\|_2 \le L_R \|v - w\|_{H^1}$ with $L_R = 1 + \max_{[-M,M]} |V''|$. Taking $w = 0$ proves membership in L^2. This argument uses a difference of the two forces; it never assigns an L^2 norm to the background separately.

For the second part, let $X_T = C([0,T]; \mathcal{E}_1)$ and put

$$(\Phi z)(t) = U_1(t) z_0 + \int_0^t U_1(t-s)(0, N_b(v(s)))\mathrm{d}s, \quad R = 2B, \quad T = \frac{1}{2L_R}.$$

Isometry and the forced estimate give

$$\|\Phi z\|_{X_T} \le B + T L_R R = R, \qquad \|\Phi z - \Phi \widetilde{z}\|_{X_T} \le \tfrac{1}{2} \|z - \widetilde{z}\|_{X_T}.$$

The closed radius-R ball is complete, so Banach's theorem gives a unique fixed point. The forcing lemma identifies it as a weak energy solution. Conversely, the force of any such solution is continuous in L^2; subtract its Duhamel construction and use L4's homogeneous uniqueness. Thus every weak energy solution is mild. Continuity puts it inside a suitable contraction ball for a short interval, giving local uniqueness; restart at each common time to obtain uniqueness on any overlap. For data in the same initial ball, subtraction gives $\|z - \widetilde{z}\|_{X_T} \le 2\|z_0 - \widetilde{z}_0\|_{\mathrm{lin}}$. Restarting also gives continuous dependence on compact intervals in a common bounded ball.

Unique solutions glue to a maximal one. If a finite endpoint T_* had a sequence $t_j \uparrow T_*$ with bounded norms, the common positive time for that ball would extend a restart beyond T_*, a contradiction. Therefore $\|z(t)\|_{\mathrm{lin}} \to \infty$ at a finite endpoint. The estimates used continuity of V'' on bounded intervals, not V'''; the assumed C^3 regularity is more than needed here.

Exercise 15.2 (The cubic survives in three dimensions). Take as given the Sobolev estimate $\|u\|_{L^6(\mathbb{R}^3)} \le C_S \|u\|_{H^1(\mathbb{R}^3)}$ for every real $u \in H^1$, where C_S is a fixed positive constant. Consider

$$u_{tt} - \Delta u + u^3 = 0, \qquad (u(0), u_t(0)) \in \mathcal{E}_3.$$

(a) Justify the three-dimensional version of the forced mass-one energy estimate, including the weak equation. Prove local existence, energy-class uniqueness and continuous dependence by a Duhamel contraction, with a common lifespan for a bounded initial ball.

(b) Would the same $H^1 \to L^2$ force argument handle u^5? For a nonzero real $\eta \in C_c^\infty(\mathbb{R}^3)$, compute the scaling of $u_\Lambda(x) = \Lambda^{1/2}\eta(\Lambda x)$, $\Lambda \ge 1$, in H^1, and of its cubic and quintic forces in L^2. Limit the conclusion to this contraction argument.

Solution. For the first part, the mass-one Fourier rotation on $(\nu\widehat{u}, \widehat{p})$ is orthogonal pointwise in $k \in \mathbb{R}^3$. Consequently U_3 is an isometry and its norm integral obeys

$$\left\| \int_0^t U_3(t-s)(0, F(s)) \mathrm{d}s \right\|_{\mathrm{lin}} \le \int_0^t \|F(s)\|_2 \mathrm{d}s.$$

The integral is a continuous energy pair. Smooth compact frequency cutoffs of $F \in C([0,T]; L^2)$ converge uniformly, by the compact-image finite-net argument of Lecture 15. On each cutoff, differentiating the sine integral gives $u_t = p$ and $p_t = \Delta u - u + F$ in L^2. The displayed bound passes pairs uniformly in energy norm to the limit; compact test functions pass the equation to the limit as well. The zero initial traces converge. L4's uniqueness identifies every forced energy solution with this construction plus its homogeneous evolution. This proves the needed dimension change without a nonlinear existence input.

Rewrite the equation as $u_{tt} - \Delta u + u = N(u) = u - u^3$. Hölder and the given embedding give, on the radius-R H^1 ball,

$$\|u^3 - v^3\|_2 \le \|u - v\|_6 (\|u\|_6^2 + \|u\|_6\|v\|_6 + \|v\|_6^2) \le 3C_S^3 R^2 \|u - v\|_{H^1}.$$

Thus $N : H^1 \to L^2$ is locally Lipschitz with $N(0) = 0$ and $L_R = 1 + 3C_S^3 R^2$. With $R = 2B$, $T = 1/(2L_R)$ the two ball estimates are $B + TL_R R = R$ and $TL_R = 1/2$. Banach's theorem constructs the mild solution. The forced identification, short-time uniqueness and overlap restart from Problem 1 give weak energy-class uniqueness, continuous dependence and the same continuation alternative.

For the second part, changing variable $y = \Lambda x$ gives

$$\|u_\Lambda\|_{H^1}^2 = \Lambda^{-2}\|\eta\|_2^2 + \|\nabla\eta\|_2^2, \quad \|u_\Lambda^3\|_2 = \|\eta^3\|_2, \quad \|u_\Lambda^5\|_2 = \Lambda\|\eta^5\|_2.$$

A locally Lipschitz map on H^1 is bounded on a sufficiently small ball about zero. Multiplying this family by any fixed small positive constant still makes the quintic norms unbounded while its H^1 norms fit that ball. Thus the quintic is not locally Lipschitz from H^1 to L^2 there. This refutes that force estimate, not local theory in other spaces or by other methods.

Exercise 15.3 (Sine-Gordon on a weighted path space). Let b be a smooth bounded stationary sine-Gordon background, $-b'' + \sin b = 0$, and write $\phi = b + v$. For any fixed $T > 0$, equip $X_T = C([0,T]; \mathcal{E}_1)$ with the equivalent norm

$$\|z\|_a = \sup_{0 \le t \le T} e^{-at} \|z(t)\|_{\mathrm{lin}}, \qquad a > 0.$$

Derive the reference mass-one Duhamel force. Show that the Duhamel map is a contraction on the entire space X_T for $a = 4$, with no radius restriction. Prove existence and uniqueness on every finite interval and give a continuous-dependence bound. Why is an amplitude-dependent local constant unnecessary? Do not assume that b or $\sin b$ belongs to L^2.

Solution. Subtracting the stationary equation gives the force

$$N_b(v) = v - \sin(b+v) + \sin b, \qquad N_b(0) = 0, \qquad \partial_v N_b(v) = 1 - \cos(b+v) \in [0,2].$$

Hence $\|N_b(v) - N_b(w)\|_2 \le 2\|v - w\|_2$, independently of the background and amplitudes. In particular the force of every continuous energy path is continuous in L^2; no separate integrability of $\sin b$ is used. The Duhamel map of Problem 1 therefore maps the entire Banach space X_T to itself. For two paths its difference obeys

$$e^{-at} \|\Phi z(t) - \Phi \widetilde{z}(t)\|_{\mathrm{lin}} \le 2\|z - \widetilde{z}\|_a \int_0^t e^{-a(t-s)} \mathrm{d}s \le \frac{2}{a} \|z - \widetilde{z}\|_a.$$

For $a = 4$ it is a strict contraction without restricting to a ball. Banach's theorem gives a unique fixed point on each prescribed finite interval. Forced identification again gives exactly the weak energy solution. Restrictions of solutions constructed for different T agree by uniqueness, so they give a global solution; the negative direction follows by time reversal. For two data, the homogeneous contribution has weighted norm at most the initial difference. Absorbing the contraction term gives weighted difference at most twice that value. Removing the weight gives

$$\sup_{[0,T]} \|z(t) - \widetilde{z}(t)\|_{\mathrm{lin}} \le 2e^{4T} \|z_0 - \widetilde{z}_0\|_{\mathrm{lin}}.$$

The global bound on cosine, rather than a supremum over an expanding field range, is what removes amplitude dependence.

Exercise 15.4 (Defocusing continuation with the analytic input stated). For the three-dimensional cubic solution of Problem 2, assume additionally that the following energy identity holds throughout its maximal energy-class lifespan:

$$E_0 = \int_{\mathbb{R}^3} \left[\frac{1}{2} u_t^2 + \frac{1}{2} |\nabla u|^2 + \frac{1}{4} u^4 \right] \mathrm{d}^3 x.$$

This is an additional hypothesis at the weak energy regularity, not a consequence of L3's classical balance. Prove that the initial integral is finite and that the solution is global. Give a bound for its full $\mathcal{E}_3$ norm on every finite time interval, explicitly recovering the L^2 displacement term.

Solution. Hölder gives $\|u_0\|_4^4 \le \|u_0\|_2 \|u_0\|_6^3 < \infty$, so the prescribed initial energy is finite. Every term in the assumed conserved energy is nonnegative. It follows that

$$\|u_t(t)\|_2 \le \sqrt{2E_0}, \qquad \|\nabla u(t)\|_2 \le \sqrt{2E_0}.$$

The energy solution has the L^2 primitive $u(t) = u_0 + \int_0^t u_t(s)\mathrm{d}s$, hence

$$\|u(t)\|_2 \le \|u_0\|_2 + |t|\sqrt{2E_0}, \qquad \|(u(t), u_t(t))\|_{\mathrm{lin}}^2 \le \left(\|u_0\|_2 + |t|\sqrt{2E_0}\right)^2 + 4E_0.$$

This stays finite on every finite time interval, contradicting the continuation alternative at any finite maximal endpoint. Global existence here follows from Problem 2 plus the explicitly assumed weak conservation identity. The argument has not proved that analytic input from the classical balance at insufficient regularity.